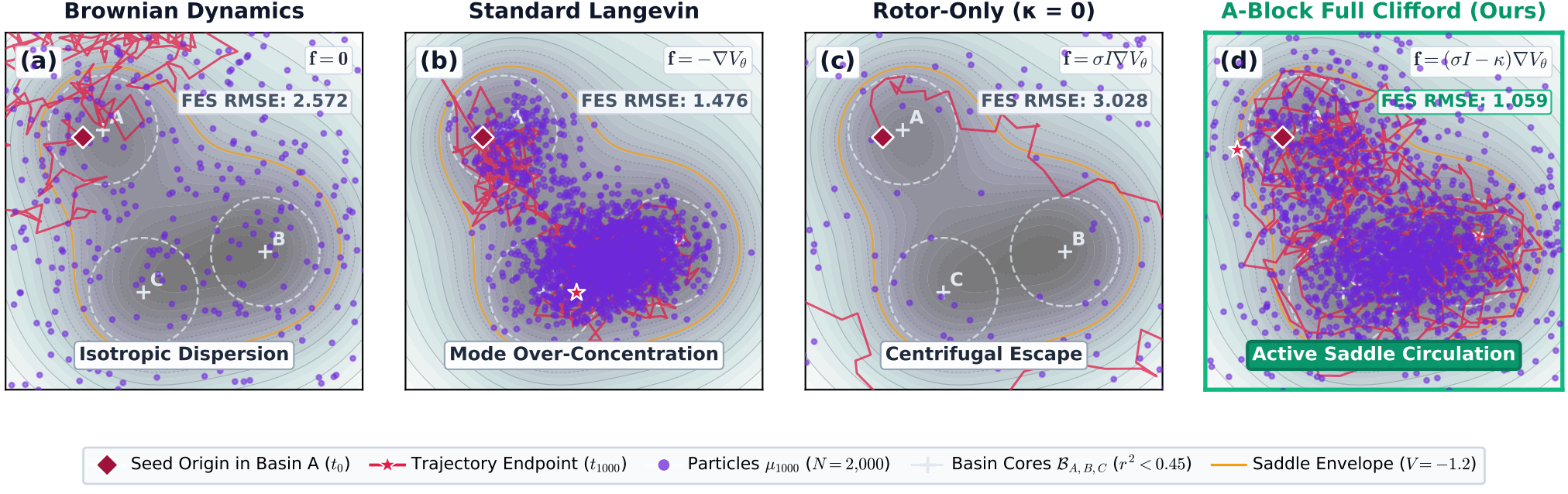

Brownian Dynamics
Standard Langevin
Rotor-Only (κ = 0)
A-Block Full Clifford (Ours)
(a)
$\mathbf{f} = 0$
FES RMSE: 2.572
Isotropic Dispersion
(b)
$\mathbf{f} = -\nabla V_\theta$
FES RMSE: 1.476
Mode Over-Concentration
(c)
$\mathbf{f} = \sigma I \nabla V_\theta$
FES RMSE: 3.028
Centrifugal Escape
(d)
$\mathbf{f} = (\sigma I - \kappa)\nabla V_\theta$
FES RMSE: 1.059
Active Saddle Circulation
A
B
C
Seed Origin in Basin A (t_0)
Trajectory Endpoint (t_{1000})
Particles μ_{1000} ($N = 2{,}000$)
Basin Cores $\mathcal{B}_{A,B,C}$ ($r^2 < 0.45$)
Saddle Envelope ($V = -1.2$)